

**STRULE AUTOMATION**

Industrial Controls · Bay Area & Northern California

TRAINING PRIMER

PID Loops Without the Math: What P, I and D Actually Do

I've lost count of how many times a maintenance lead has walked me over to a human-machine interface (HMI) screen, pointed at a level that keeps wandering up and down, and said "the loop won't hold." Nine times out of ten the person asking is sharp, knows the machine cold, and just never had anybody explain what those three numbers in the controller are really doing. So that's what these field notes are for. No calculus, no Laplace transforms, no textbook. Just the picture in my head when I look at a trend and decide what to touch. PID stands for proportional-integral-derivative, and by the end of this you'll know what each of those three does, what it looks like on a trend when it's wrong, and how to nudge one safely without lighting anything up.

Why this matters on the floor

A control loop is the thing that keeps a number where you want it without a human standing there turning a valve. Tank level, discharge pressure, a temperature, a flow. When it's tuned right, you forget it exists. When it's tuned wrong, it either sits stubbornly off-target, or it hunts up and down forever, and pretty soon somebody's chasing a "process problem" that's really just a controller that needs a better setting. Knowing the difference saves you hours and keeps you from blaming the wrong part.

The pieces of a feedback loop

Let me set up one concrete example and use it the whole way through. Say you've got a surge tank or a supply tank, and a feed pump driven by a variable frequency drive (VFD) — that's the electronic box that changes motor speed by changing the frequency of the power feeding it. The pump fills the tank; downstream demand pulls water back out. You want to hold the tank level steady at, say, the halfway mark, because the downstream draw keeps changing and you don't want the tank to overflow or run dry. When the level drops, you speed the pump up to fill; when it climbs, you ease off.

Four terms, and they're all you need:

- **Setpoint (SP):** where you want the level to be. Halfway. This is what you tell the controller.
- **Process variable (PV):** where the level actually is right now, as read by the level transmitter — the sensor that turns real tank level into an electrical signal the PLC can read. PLC means programmable logic controller, the industrial computer running the loop.

- **Controller output (CO):** the command the controller sends out. In our case it's a speed reference to the VFD, usually thought of as 0 to 100 percent.
- **Error:** the gap between where you want to be and where you are. Plainly, $\text{error} = \text{SP} - \text{PV}$. If the setpoint is 50 percent and the tank is at 45 percent, your error is 5 percent, and the controller needs to do something about it.

That's the whole game. The controller looks at the error, decides how hard to run the pump, the level moves, it reads the new error, and it does it again, many times a second. PID is just the recipe for turning that error into an output. Three ingredients. Let's take them one at a time.

Proportional: react to how big the error is right now

Proportional (the P) is the muscle. It pushes in proportion to the size of the error this instant. Big error, big push. Small error, small push. If the tank is way below setpoint, P slams the pump command hard. As the level climbs and the error shrinks, P eases off on its own.

Turn P up and the controller reacts harder and faster to any gap. Turn it up too far and you get the classic problem: it overshoots, then overcorrects the other way, then overcorrects again, and now the level is cycling up and down — oscillation that P itself is feeding. Turn P down and the loop gets gentle and lazy: it takes forever to respond, and it tends to stall out before it ever reaches setpoint.

That last part trips people up, so sit with it. With P only, the controller needs *some* error to produce *some* output. If the tank is exactly at setpoint, error is zero, so the proportional push is zero — and if you need the pump running at 40 percent just to keep up with what's being drawn off downstream, a zero push won't hold it there. So the level settles at a spot a little off from setpoint, where the leftover error produces exactly the push you need. That permanent gap has a name: **steady-state offset** (also called droop). Proportional-only loops almost always live with a little offset. That is simply what P does when it works alone, and it is exactly why we bring in the second ingredient.

Integral: erase the leftover offset over time

Integral (the I) is patient. It watches the error and, as long as any error is still hanging around, it keeps slowly winding the output further in the correcting direction. It doesn't care how big the error is this instant — it cares that the error hasn't gone away yet. Give it time and it walks the output to wherever it needs to be to drive the error all the way to zero. That's the term that kills the steady-state offset P leaves behind.

The trade is speed versus stability. Turn I up (in most controllers that means a shorter integral time, so watch which direction your particular box uses) and it erases offset faster — but wind it too aggressively and it overshoots past setpoint, then has to unwind, and you get a slow, rolling oscillation — a long, lazy wallow that can take a minute or more per cycle, nothing like the fast buzz too much P gives you. Turn I too weak and offset just sits there for a long time, creeping toward setpoint so slowly nobody believes it'll ever arrive.

For our tank: P gets the level back in the neighborhood quickly, and I quietly trims the pump speed until the level parks right on 50 percent and stays. P plus I is the workhorse combination. The large majority of loops I touch in aggregate plants, material recovery facilities (MRFs), and organics or anaerobic digestion (AD) sites run just fine on P and I alone.

Derivative: react to how fast the error is changing

Derivative (the D) watches the *rate* of change. Not how big the error is, not how long it's been there, but how fast it's moving. If the level is diving toward setpoint quickly, D sees that speed and starts easing off early, so you don't blow past the target. Used well, it acts like a shock absorber that takes the edge off overshoot.

Here's the catch, and it's a big one on our kind of signals: D reacts to *change*, and electrical noise is nothing but rapid change. A level or flow signal bouncing a percent or two from splashing, turbulence, or plain electrical hash looks to the derivative term like the error is whipping back and forth violently. D dutifully amplifies that and jitters the output, which can drive the VFD nuts and wear things out. That's why an awful lot of real-world loops — especially noisy level and flow loops — run with D set to zero and are better for it. I leave D off by default and only reach for it on slower, cleaner loops like temperature, where change is gradual and the signal is smooth. If you're staring at a loop wondering whether to use D, the safe answer on a noisy plant signal is usually no.

Reading the failure on a trend

Pull up a trend of PV against SP — most HMIs and historians (the software that logs and stores your process values over time so you can look back at them) will overlay them — and the shape tells you which ingredient is off. Watch a few minutes before you touch anything.

- **A steady gap that never closes.** PV settles and holds a fixed distance from SP. That's steady-state offset. You've got P doing its job but not enough I (or none) to erase the leftover. Add or strengthen integral.
- **Slow, rolling cycling — long waves, maybe a minute or more each.** Too much integral. The loop overshoots, unwinds, overshoots. Back the I off.
- **Fast, tight cycling — a rapid buzz around setpoint.** Too much proportional (or, less often, D amplifying noise). Back the P off first.
- **Sluggish — takes forever to react, drifts toward setpoint at a crawl.** Not enough P, and probably not enough I either. It's stable but half asleep. Nudge P up a little.

One honest warning before you start turning knobs: some loops "won't tune" because the problem isn't in the controller at all. A valve that's sticking, a pump that's oversized or undersized for the duty, a check valve chattering, a transmitter reading garbage, a tank so small that any pump move swings the level hard — none of that gets fixed by touching P, I, or D. If you've made reasonable moves and the

loop still fights you, step back and ask whether you're looking at a mechanical or a sizing problem wearing a tuning costume. I've watched people spend a full shift retuning around a valve that just needed rebuilding.

Try this on your own plant, safely

Pick one loop that's misbehaving in a way you can tolerate experimenting with, and never one tied to a safety function. Then:

1. **Trend first.** Put PV and SP on one pen and just watch. Name the shape from the list above before you change anything. Half the job is diagnosis.
2. **Start conservative and go slow.** If you're not sure, gentler is safer — a soft loop drifts; an aggressive one can slam a pump or cycle a valve hard enough to matter.
3. **One change at a time, and write it down.** Change P, or change I, never both at once, or you'll never know which move did what. Log the old value, the new value, the date, and who made it. That log is management-of-change, and it's how the next person (or you, in six months) understands the machine.
4. **Watch a full trend after each change** before deciding the next one. Give the loop time to show you its new behavior.
5. **Stay inside the equipment's rated limits.** Tune the loop; don't reprogram around a protection. Never disable, bypass, or defeat a high-level trip, a pressure relief, a motor overload, or any interlock (protective logic that stops or blocks an action to keep equipment or people safe) to make a loop look better. If a genuine nuisance setpoint is tripping inside a safe range, the legitimate path is a logged, single-change management-of-change to retune that one setpoint within the rated limit — not ripping the protection out.

A word on the physical side, because tuning sometimes has you near live gear. If you end up taking a reading on energized equipment to chase a signal problem, that's arc-flash territory: right personal protective equipment (PPE), rated meter and leads, and treat it as the hazard it is. On a VFD, remember the DC bus holds a charge after you kill power — the drive can bite you for several minutes after the disconnect is open, so wait out the drive's stated discharge time and verify before you reach in. Any hands-on mechanical work — pulling a valve, touching a pump coupling — is lockout/tagout (LOTO) and verified zero energy first, every time, no exceptions. And if this loop lives on a digester or anywhere with biogas, don't forget what's in the air: hydrogen sulfide (H₂S) is immediately dangerous to life or health (IDLH) around 100 parts per million (ppm) and deadens your sense of smell so you can't trust your nose, methane is flammable between roughly 5 and 15 percent in air (its lower and upper explosive limits, LEL and UEL), those spaces are permit-required confined spaces, you wear a calibrated 4-gas monitor, and the area is Class I hazardous-location rated under the National Electrical Code (NEC 500/505). None of that is optional.

One last thing on forcing. If you force a PLC input or output to test a loop — jamming a value to see how the system responds — understand you've left a live plant in a non-normal state where the controller no longer sees reality. That's a real diagnostic tool and sometimes the right call, but a force is never something you walk away from. Clear it before you leave, and if you can't finish, tell the next shift in plain words what's forced and why.

| The short version

Proportional is the muscle that pushes on the error you have right now. Integral is the patience that erases the small leftover gap over time. Derivative is the shock absorber that watches how fast things are moving, and on noisy plant signals it usually causes more trouble than it's worth, so most loops leave it off. Get comfortable reading the shape of a trend, make one logged change at a time inside the rated limits, and keep in the back of your mind that a loop which won't behave might be pointing at a valve or a pump rather than a number.

Where I see this go sideways most is on plants where the controls have been handed between vendors over the years and nobody's sure who set what. A second set of eyes on the tuning and the trend history usually pays for itself there — not because the loop is exotic, but because the reasoning behind the settings walked out the door with the last integrator.

| Glossary

- **PID (proportional-integral-derivative):** the three-part recipe a controller uses to turn error into an output.
- **VFD (variable frequency drive):** electronic drive that sets motor speed by varying the frequency of its output power.
- **PLC (programmable logic controller):** the industrial computer that runs the control loop.
- **HMI (human-machine interface):** the operator screen showing values and trends.
- **Historian:** software that logs and stores process values over time so you can look back at them.
- **Interlock:** protective logic that stops or blocks an action to keep equipment or people safe.
- **Setpoint (SP):** the target value you want the process to hold.
- **Process variable (PV):** the actual measured value right now, from the transmitter.
- **Controller output (CO):** the command the controller sends out, such as a speed reference to a VFD.
- **Error:** the gap between target and actual, SP minus PV.
- **Steady-state offset (droop):** the permanent small gap a proportional-only loop settles at.
- **Proportional (P):** control action sized to how big the error is right now.
- **Integral (I):** control action that keeps correcting as long as any error remains, erasing offset over time.

- **Derivative (D):** control action based on how fast the error is changing; amplifies noise.
- **MRF (material recovery facility):** a plant that sorts and recovers recyclables.
- **AD (anaerobic digestion):** the process that breaks down organics without oxygen and produces biogas.
- **Management-of-change:** the practice of making one deliberate, logged change with a record of what and why.
- **LOTO (lockout/tagout):** isolating and verifying zero energy before hands-on work.
- **PPE (personal protective equipment):** protective gear, including arc-flash-rated gear for live electrical work.
- **DC bus:** the internal capacitor bank in a drive that holds a charge after power is removed.
- **H₂S (hydrogen sulfide):** toxic gas in biogas, IDLH near 100 ppm, deadens the sense of smell.
- **LEL / UEL (lower / upper explosive limit):** the concentration range in which a gas will burn; methane roughly 5 to 15 percent in air.
- **IDLH (immediately dangerous to life or health):** the exposure level at which escape becomes life-threatening.
- **NEC 500/505:** National Electrical Code sections defining hazardous-location electrical requirements.